

PLASTIBIOFUEL LLC · HASLET, TEXAS

Post-consumer PET to chemicals

One page, written to be forwarded. Every number on it traces to a signed independent report.

122 MICROMOL/H/MG TURNOVER, IMMOBILIZED	3 MILESTONES CLOSED EARLY	1 PROVISIONAL PATENT
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ENTITY PlastiBioFuel LLC. UEI KQAWZ54RDUM8, CAGE 175D6. Service-disabled veteran-owned small business, certified and active in SAM.gov.	NAICS 325199 other basic organic chemical manufacturing; 325193 ethyl alcohol manufacturing; 541715 research and development.
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WHAT WE DO A PET hydrolase immobilized in a covalent organic framework depolymerizes post-consumer PET. A continuous spiral-flow reactor, designed in simulation, is the intended production form.	RESEARCH PARTNER University of North Texas, Prof. Shengqian Ma, under Sponsored Research Agreement SRA1016. Framework and immobilization chemistry.
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INTELLECTUAL PROPERTY USPTO provisional application 63/848,456, filed July 2025. Framework immobilization, spiral-flow reactor geometry, and the integrated process.
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ESTABLISHED BY INDEPENDENT REPORT	UNT Final Project Report, Phase I, signed 2026-08-07
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Depolymerization confirmed	Free FAST-PETase breaks real post-consumer water bottle PET down to terephthalic acid, confirmed by HPLC against standards.
Immobilization confirmed	The enzyme loads into a TAPB-BPDA-COF with no protein detectable in the supernatant at 280 nm.
Activity retained	Immobilized turnover measured 122 micromol per hour per milligram at 24 hours, against a 246 to 348 literature range for the free enzyme.

Bottleneck identified	The laboratory attributes that gap to mass transfer down the framework channel rather than to loss of enzyme activity. Transport is what reactor geometry decides.
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Schedule	Three milestones completed ahead of schedule.
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NOT YET ESTABLISHED, STATED PLAINLY

No reactor built	The spiral-flow reactor exists as a validated model. The report lists vessel shape and size as future work.
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No ethanol produced	The route from monomer to fuel-grade ethanol is the objective and has not been demonstrated.
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No offtake executed	Commercial offtake is in discussion. There is no signed letter of intent and no executed agreement.
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Fuel credits unqualified	Any renewable fuel credit eligibility remains subject to final pathway qualification.
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WHAT WE ARE LOOKING FOR

Partners and programs in chemicals from waste feedstocks, pre-pilot reactor capacity, and downstream separations. First-time federal applicant, prepared to work as prime or subrecipient.

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